University of Vienna

Faculty of Mathematics

Exam of Linear Algebra in Data Science (250076 VO)

Tuesday July 7, 2026, 15:00 - 16:30

- Write clearly and motivate all your answers.
- The maximal number of points in the exam is 24. The grade is then transformed into the official system of points ranging from 1 (perfect score of 24/24 points) to 6 (worst score of 0/24 points).

Question 1 (**Stable subspaces - 5 points**).
Let E be a $\mathbb{K}$-vector space, let $f \in \mathcal{L}(E)$ and let $p \in \mathcal{L}(E)$ be a projection operator. Show that

$$p \circ f = f \circ p \quad \Longleftrightarrow \quad \operatorname{im}(p) \text{ and } \ker(p) \text{ are stable by } f\ .$$

Solution.

($\Rightarrow$) Suppose that p and f commute, then:

- for any $y \in \operatorname{im} p$, we have $x \in E$ such that $p(x) = y$, we thus have

$$p[f(x)] = f[\underbrace{p(x)}_{=y}] = f(y) \quad \Rightarrow \quad f(y) \in \operatorname{im} p,$$

  which proves that $f(\operatorname{im} p) \subset \operatorname{im} p$, i.e. $\operatorname{im} p$ is stable by f.
- for any $x \in \ker p$, we have

$$p[f(x)] = f[\underbrace{p(x)}_{=0}] = f(0) = 0 \quad \Rightarrow \quad f(x) \in \ker p,$$

  which proves that $f(\ker p) \subset \ker p$, i.e. $\ker p$ is stable by f.

($\Leftarrow$) Suppose now that $\operatorname{im} p$ and $\ker p$ are stable by f, i.e.

$$f(\operatorname{im} p) \subset \operatorname{im} p, \quad \text{and} \qquad (1) \quad \{exam:stablei$$
$$f(\ker p) \subset \ker p. \qquad (2) \quad \{exam:stablek$$

recall that for projections p, we have the following properties:

xam:directsum (A) E can be decomposed in the following direct sum:

$$E = \operatorname{im} p \oplus \ker p.$$

:projectionid (B) p restricted on the subspace $\operatorname{im} p$ is the identity, i.e.

$$p\big|_{\operatorname{im} p} = \mathrm{Id}_{\operatorname{im} p}.$$

We now define the *commutator* of f and p:

$$g = [f, g] = f \circ p - p \circ f.$$

Note that if $g \in \mathcal{L}(E)$ is the zero mapping, then f and p commute.
Let $x \in E$, by (A) there exists an unique decomposition $x = y + z$, with $y \in \operatorname{im} p$ and $z \in \ker p$. We thus have

$$\begin{aligned} g(x) = g(y+z) &= f[\underbrace{p(y)}_{\overset{(B)}{=} y}] + f[\ \underbrace{p(z)}_{\overset{(z \in \ker p)}{=} 0}\ ] - \underbrace{p[f(y)]}_{\overset{(1),(B)}{=} f(y)} - \underbrace{p[f(z)]}_{\overset{(2)}{=} 0} \\ &= f(y) - f(y) = 0. \end{aligned}$$

Since the above equation holds for all $x \in E$, we have that $g = 0$.

Question 2 (**Diagonalization - 11 points**).
Let E be an n-dimensional vector space and $f \in \mathcal{L}(E)$.

1. Define the concept of the matrix representation of f. **(2 points)**
2. State the theorem providing the three necessary and sufficient conditions for f to be diagonalizable. **(2 points)**

In the rest of the exercise, $E = \mathbb{R}^4$ and we consider the endomorphism $f \in \mathcal{L}(\mathbb{R}^4)$ whose matrix representation in the canonical basis $\mathcal{E}$ is

$$[f]_{\mathcal{E}}^{\mathcal{E}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ h & h & 0 & 0 \\ 0 & 0 & h & h \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \text{for all} \quad h \in \mathbb{R}^*. \tag{3}$$

The goal of the exercise is to study the diagonalizability of f depending on h when $h \in \mathbb{R}^*$. For this:

3. Compute the characteristic polynomial of f, and present the result in a factorized form. **(2 points)**
4. Deduce its eigenvalues and its algebraic multiplicities. **(1 point)**
5. Compute the eigenspaces, and give the geometric multiplicities. **(3 points)**
6. For which values of $h \in \mathbb{R}^*$ is f diagonalizable? **(1 point)**

Solution. [**Theoretical parts 1 and 2 skipped**]

1. See Eq. (1.3) in the lecture notes (version of June 29, 2026)
2. See Theorem 5.20 in the lecture notes (version of June 29, 2026).
3. The characteristic polynomial can be obtained in the following way:

$$\begin{aligned} p_f(t) &= \det\left([f]_{\mathcal{E}}^{\mathcal{E}} - t \cdot \mathrm{Id}\right) = \det \begin{pmatrix} 1-t & 0 & & \\ h & h-t & & \\ & & h-t & h \\ & & 0 & 1-t \end{pmatrix} \\ &= \det \begin{pmatrix} 1-t & 0 \\ h & h-t \end{pmatrix} \cdot \det \begin{pmatrix} h-t & h \\ 0 & 1-t \end{pmatrix} = (1-t)^2(h-t)^2 \end{aligned}$$

4. Depending on the values of h, we have the following cases:

($h = 1$) The only eigenvalue of f is $\lambda_0 = 1$, with algebraic multiplicity 4.

($h \neq 1$) We have two eigenvalues $\lambda_1 = 1$ and $\lambda_2 = h$. Each with algebraic multiplicity 2.

5. We shall distinguish the different cases of h, the same way as the previous part:

($h = 1$) We have that

$$F_{\lambda_0} = \ker\left([f]_{\mathcal{E}}^{\mathcal{E}} - \lambda_0 \cdot \mathrm{Id}\right) = \ker \begin{pmatrix} 0 & 0 & & \\ 1 & 0 & & \\ & & 0 & 1 \\ & & 0 & 0 \end{pmatrix} = \operatorname{span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

where we see that $\lambda_0 = 1$ has geometric multiplicity 2.

($h \neq 1$) in this case we have

$$F_{\lambda_1} = \ker\left([f]_{\mathcal{E}}^{\mathcal{E}} - \lambda_1 \cdot \mathrm{Id}\right) = \ker \begin{pmatrix} 0 & 0 & & \\ h & h-1 & & \\ & & h-1 & h \\ & & 0 & 0 \end{pmatrix} = \operatorname{span}\left\{ \begin{pmatrix} 1-h \\ h \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ h \\ 1-h \end{pmatrix} \right\},$$

$$F_{\lambda_2} = \ker\left([f]_{\mathcal{E}}^{\mathcal{E}} - \lambda_2 \cdot \mathrm{Id}\right) = \ker \begin{pmatrix} 1-h & 0 & & \\ h & 0 & & \\ & & 0 & h \\ & & 0 & 1-h \end{pmatrix} = \operatorname{span}\left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Where we see that both eigenvalues λ_1 and λ_2 have geometric multiplicity 2.

6. Since the only case in which the geometric and algebraic multiplicities of the eigenvalue(s) agree is when $h \neq 1$, we have that in this case f is diagonalizable.

Question 3 (**Decomposition - 8 points**).
Let $(E, \langle \cdot, \cdot \rangle)$ be an n-dimensional $\mathbb{K}$-inner product space with $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. The space has inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|\cdot\|$. Let u be a bijective endomorphism. The goal of this exercise is to prove that there exist:

- an isometry $\sigma \in \mathcal{L}(E)$, which means that it is such that
$$\|\sigma(x)\| = \|x\|, \quad \forall x \in E,$$
- a self-adjoint endomorphism $\tau \in \mathcal{L}(E)$ with strictly positive eigenvalues,

such that $u = \sigma \circ \tau$.

1. Justify that the mapping $w := u^* \circ u$ is diagonalizable in an orthonormal basis of E (which we will call $\mathcal{E} = \{e_1, \dots, e_n\}$). Show that its eigenvalues are real and strictly positive. **(2 points)**
2. In the following, we denote by λ_i the eigenvalue of w associated to the eigenvector e_i for $i \in \{1, \dots, n\}$. Express $\|u(e_i)\|$ as a function of λ_i. **(1 point)**
3. Show that the endomorphism $\sigma \in \mathcal{L}(E)$ defined as
$$\sigma(e_i) := \frac{u(e_i)}{\|u(e_i)\|}, \quad \forall i \in \{1, \dots, n\}, \tag{4}$$
is an isometry. **(3 points)**
4. We define $\tau = \sigma^{-1} \circ u$. Show that
$$\tau(e_i) = \lambda_i^{1/2} e_i, \tag{5}$$
and deduce from this that τ is self-adjoint, with strictly positive eigenvalues. **(2 points)**

Solution.

1. w is an self adjoint operator:
$$w^* = (u^* \circ u)^* = u^* \circ (u^*)^* = u^* \circ u = w,$$
where in the penultimate equality we've used that $(f^*)^* = f$ for any mapping $f : E \to E$ where E is a finite dimensional vector space. Therefore by the *Spectral Theorem* there exists an orthonormal basis $\mathcal{E}$ (with respect to the inner product $\langle \cdot, \cdot \rangle$) that diagonalizes the operator w. Furthermore, since w is self adjoint, the eigenvalues of w satisfy $\lambda_i \in \mathbb{R}$.
To see that $\lambda_i > 0$, we first show that w is strictly positive, but this is indeed true since for each $x \in E \backslash \{0\}$ we have
$$\langle w(x), x \rangle = \langle u^* \circ u(x), x \rangle = \langle u(x), u(x) \rangle = \|u(x)\|^2 > 0, \tag{6}$$ {exam:strictl
where in the last equality we have used that $u(x) \neq 0$, $\forall x \in E \backslash \{0\}$ (u is bijective). Since if (λ_i, e_i) is an eigenpair of w, then
$$0 \overset{(6)}{<} \langle \underbrace{w(e_i)}_{=\lambda_i e_i}, e_i \rangle = \lambda_i \underbrace{\|e_i\|^2}_{>0} \quad \Rightarrow \quad \lambda_i = \frac{\langle w(e_i), e_i \rangle}{\|e_i\|^2} > 0.$$
2. Since e_i are eigenvectors of w with eigenvalue λ_i, we see that
$$\|u(e_i)\|^2 = \langle u(e_i), u(e_i) \rangle = \langle u^* \circ u(e_i), e_i \rangle = \langle w(e_i), e_i \rangle = \langle \lambda_i e_i, e_i \rangle = \lambda_i \underbrace{\|e_i\|^2}_{=1} = \lambda_i,$$
which means that $\|u(e_i)\| = \sqrt{\lambda_i}$.

3. Let $x \in E$, we decompose x as linear combination of the basis vectors e_i:

$$x = \sum_{i=1}^{n} \mu_i e_i, \text{ where } \mu_i \in \mathbb{K},$$

We have that

$$\|x\|^2 = \sum_{i=1}^{n} |\mu_i|^2 \underbrace{\|e_i\|^2}_{=1} + 2 \sum_{1 \le i < j \le n} \mu_i \bar{\mu}_j \underbrace{\langle e_i, e_j \rangle}_{=0} = \sum_{i=1}^{n} |\mu_i|^2 \tag{7}$$ {exam:xnorm}

Moreover, we have that

$$\sigma(x) = \sum_{i=1}^{n} \mu_i \sigma(e_i) = \sum_{i=1}^{n} \mu_i \frac{u(e_i)}{\|u(e_i)\|},$$

taking the square of the norm of $\sigma(x)$ yields:

$$\begin{aligned}
\|\sigma(x)\|^2 &= \sum_{i=1}^{n} |\mu_i|^2 \frac{\|u(e_i)\|^2}{\|u(e_i)\|^2} + 2 \sum_{1 \le i < j \le n} \frac{\mu_i \bar{\mu}_j}{\|u(e_i)\| \|e_j\|} \langle u(e_i), u(e_j) \rangle \\
&= \sum_{i=1}^{n} |\mu_i|^2 + 2 \sum_{1 \le i < j \le n} \frac{\mu_i \bar{\mu}_j}{\|u(e_i)\| \|e_j\|} \underbrace{\langle u^* \circ u(e_i), e_j \rangle}_{=\langle w(e_i), e_j \rangle} \\
&= \sum_{i=1}^{n} |\mu_i|^2 + 2 \sum_{1 \le i < j \le n} \frac{\mu_i \bar{\mu}_j}{\|u(e_i)\| \|e_j\|} \langle \lambda_i e_i, e_j \rangle \\
&= \sum_{i=1}^{n} |\mu_i|^2 + 2 \sum_{1 \le i < j \le n} \frac{\mu_i \bar{\mu}_j \lambda_i}{\|u(e_i)\| \|e_j\|} \underbrace{\langle e_i, e_j \rangle}_{=0} \\
&= \sum_{i=1}^{n} |\mu_i|^2 \overset{(7)}{=} \|x\|^2,
\end{aligned}$$

where we see that $\|\sigma(x)\| = \|x\|$, or, in other words, σ is an isometry.

4. We have proven that σ is an isometry, therefore σ has to be a bijective endomorphism, which means that σ^{-1} is well defined. For each $i \in \{1, \ldots, n\}$, we have

$$\sigma(e_i) = \frac{u(e_i)}{\|u(e_i)\|} \iff \sigma^{-1}\left(\frac{u(e_i)}{\|u(e_i)\|}\right) = e_i \iff \sigma^{-1} \circ u(e_i) = \|u(e_i)\| \cdot e_i = \sqrt{\lambda_i} \cdot e_i.$$

Where in the second equivalence we have used the fact that σ^{-1} exists and is well defined.
From the previous equation we see that $0 < \sqrt{\lambda_i}$ are eigenvalues of τ, with associated eigenbasis $\mathcal{E}$. Furthermore, since:

$$\langle \tau e_i, e_i \rangle = \sqrt{\lambda_i} \langle e_i, e_i \rangle = \langle e_i, \tau e_i \rangle, \quad \forall\, i \in \{1, \ldots, n\}$$

and

$$\langle \tau e_i, e_j \rangle = \sqrt{\lambda_i} \underbrace{\langle e_i, e_j \rangle}_{=0} = \sqrt{\lambda_j} \underbrace{\langle e_i, e_j \rangle}_{=0} = \langle e_i, \tau e_j \rangle, \quad \forall\, i \neq j,$$

we deduce that τ is self adjoint.